

SIGNALS

1.Input:

```
sushmitha@LAPTOP-B907B0  ×  +  ∨
GNU nano 8.7.1                signal.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
void handler(int sig)
{
printf("\nSIGINT received!\n");
printf("Signal number = %d\n", sig);
}
int main()
{
signal(SIGINT, handler);
while (1)
{
printf("Program is running...\n");
sleep(2);
}
return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano signal.c
sushmitha@LAPTOP-B907B06L:~$ ./signal
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
Signal number = 2
Program is running...
Program is running...
^Z
[3]+  Stopped                  ./signal
sushmitha@LAPTOP-B907B06L:~$ |
```

2.Input:

```
sushmitha@LAPTOP-B9O7BO  ×  +  ▾
GNU nano 8.7.1 sigterm.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if (sig == SIGTERM)
        printf("\nSIGTERM received!\n");
}

int main()
{
    signal(SIGTERM, handler);

    printf("Process ID: %d\n", getpid());
    printf("Waiting for SIGTERM...\n");

    while (1)
    {
        sleep(2);
    }

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B9O7B06L:~$ nano sigterm.c
sushmitha@LAPTOP-B9O7B06L:~$ gcc sigterm.c -o sigterm
sushmitha@LAPTOP-B9O7B06L:~$ ./sigterm
Process ID: 761
Waiting for SIGTERM...
761

SIGTERM received!
^C
sushmitha@LAPTOP-B9O7B06L:~$ |
```

3.Input:

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>

void child_handler(int sig)
{
    printf("SIGCHLD received: Child has terminated\n");
    wait(NULL);
}

int main()
{
    signal(SIGCHLD, child_handler);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child is running...\n");
        sleep(2);
        printf("Child is exiting...\n");
    }
    else
    {
        printf("Parent is waiting...\n");
        sleep(5);
        printf("Parent finished.\n");
    }

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano sigchild.c
sushmitha@LAPTOP-B907B06L:~$ nano sigchild.c
sushmitha@LAPTOP-B907B06L:~$ gcc sigchild.c -o sigchild
sushmitha@LAPTOP-B907B06L:~$ ./sigchild
Parent is waiting...
Child is running...
Child is exiting...
SIGCHLD received: Child has terminated
Parent finished.
sushmitha@LAPTOP-B907B06L:~$ |
```

4.Input:

```
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigpipe(int sig)
{
    printf("\nSIGPIPE received!\n");
    printf("The reading end of the pipe is closed.\n");
    exit(1);
}

int main()
{
    int fd[2];

    signal(SIGPIPE, handle_sigpipe);

    pipe(fd);

    printf("Pipe created.\n");

    // Close the reading end
    close(fd[0]);

    printf("Reading end closed.\n");

    printf("Trying to write to the pipe...\n");

    write(fd[1], "Hello", 5);

    printf("Write completed.\n");

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano sigpipe.c
sushmitha@LAPTOP-B907B06L:~$ gcc sigpipe.c -o sigpipe
sushmitha@LAPTOP-B907B06L:~$ ./sigpipe
Pipe created.
Reading end closed.
Trying to write to the pipe...

SIGPIPE received!
The reading end of the pipe is closed.
sushmitha@LAPTOP-B907B06L:~$ |
```

5.Input:

```
#include <stdio.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigsegv(int sig)
{
    printf("\nSIGSEGV received!\n");
    printf("Invalid memory access detected.\n");
    exit(1);
}

int main()
{
    signal(SIGSEGV, handle_sigsegv);

    printf("Program started.\n");

    int *p = NULL;

    printf("Trying to access invalid memory...\n");

    *p = 10;

    printf("Program completed.\n");

    return 0;
}
```

Output:

```
sushmitha@LAPTOP-B907B06L:~$ nano sigsegv.c
sushmitha@LAPTOP-B907B06L:~$ gcc sigsegv.c -o sigsegv
sushmitha@LAPTOP-B907B06L:~$ ./sigsegv
Program started.
Trying to access invalid memory...

SIGSEGV received!
Invalid memory access detected.
sushmitha@LAPTOP-B907B06L:~$ |
```